

# Notes: French and Poterba (AER, 1991)

## Investor Diversification and International Equity Markets

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# 1 Big picture

- **Question.** Why do investors hold such a large fraction of their equity wealth in domestic stocks, even though international equity markets offer meaningful diversification benefits?
- **Object of interest.** The paper studies the international equity portfolio weights of U.S., Japanese, and British investors across the six largest equity markets: the United States, Japan, the United Kingdom, France, Germany, and Canada.
- **Empirical fact.** Investors are heavily home-biased. In December 1989, U.S. investors held about 93.8% of their equity portfolios in U.S. stocks, Japanese investors held about 98.1% in Japanese stocks, and British investors held about 82.0% in U.K. stocks.
- **Puzzle.** Return correlations across national equity markets are far below one. For the 1975–1989 period, the average pairwise correlation among quarterly real equity returns in the six countries is about 0.502. This means international diversification should reduce portfolio risk.
- **Method.** The authors use a simple mean–variance/exponential-utility model to infer the expected return beliefs that would make observed home-biased portfolios optimal.
- **Revealed-preference logic.** If investors knowingly choose extremely domestic portfolios, then either foreign assets must look unattractive to them, domestic assets must look especially attractive, foreign risks must be perceived as larger, or some constraint must prevent foreign investment.
- **Main quantitative result.** Actual portfolio allocations imply that investors in each country expect their domestic market to earn returns several hundred basis points higher than many foreign markets.
- **Benchmark result.** Relative to a global value-weighted portfolio, observed portfolios can be rationalized by domestic “optimism” of roughly 90 basis points for U.S. investors, 250 basis points for Japanese investors, and more than 400 basis points for British investors.
- **Institutional explanation.** The paper argues that taxes, transaction costs, and formal investment limits are unlikely to fully explain the lack of international diversification, especially because the observed level of foreign investment lies below plausible regulatory ceilings.
- **Behavioral explanation.** A more plausible interpretation is that investors are optimistic about familiar domestic markets or perceive unfamiliar foreign markets as riskier than their statistical return volatility would suggest.
- **Contribution.** The paper provides an early quantitative statement of the international equity home-bias puzzle: investors appear to forgo diversification benefits because of domestic preference, beliefs, or familiarity, not simply because foreign investment is impossible.
- **Takeaway.** Home bias is not a small portfolio nuisance. In a standard portfolio model, the observed allocations require large wedges in expected returns or perceived risks across otherwise accessible national equity markets.

## 2 Data and measurement (key objects)

## 2.1 Why this setting is useful

- International equity markets are a clean place to study underdiversification because the diversification motive is transparent: national equity returns are not perfectly correlated.
- The setting allows the authors to compare actual portfolio weights with a simple benchmark: a globally diversified, value-weighted equity portfolio.
- Because the paper uses the portfolio choices of investors from different countries, it can ask whether the same asset is valued differently by domestic and foreign investors.

**Interpretation.** The paper does not estimate a causal effect. Instead, it uses observed portfolio choices to infer the beliefs or frictions that would be needed to rationalize those choices.

## 2.2 Country coverage

The analysis focuses on the six largest equity markets in the late 1980s:

$$\mathcal{C} = \{\text{US, Japan, UK, France, Germany, Canada}\}. \quad (1)$$

These markets are used for two reasons:

- they account for most world equity market capitalization;
- they have enough return and ownership data to construct portfolio weights and covariance estimates.

**Interpretation.** The six-country universe is not the full world portfolio, but it captures the main developed equity markets relevant for large international investors at the time.

## 2.3 Portfolio weights

For each investor country  $j$  and equity market  $i$ , define:

$$w_{ij} = \frac{\text{equity wealth of investors from country } j \text{ invested in market } i}{\text{total equity wealth of investors from country } j \text{ invested in the six markets}}. \quad (2)$$

The main empirical object is therefore the country-specific portfolio vector:

$$w_j = (w_{\text{US},j}, w_{\text{Japan},j}, w_{\text{UK},j}, w_{\text{France},j}, w_{\text{Germany},j}, w_{\text{Canada},j})'. \quad (3)$$

The key observed domestic weights are:

$$w_{\text{US,US}} = 0.938, \quad w_{\text{Japan,Japan}} = 0.9811, \quad w_{\text{UK,UK}} = 0.820. \quad (4)$$

**Interpretation.** A domestic equity weight near one is the empirical signature of home bias. The important question is whether such weights can be rationalized by plausible expected-return differences, risks, taxes, or constraints.

## 2.4 Data sources for ownership and market values

The portfolio estimates are described as crude but informative. They are based on:

- tabulations using the *U.S. Treasury Bulletin*;
- international equity flow and ownership data from Howell and Cozzini;
- Morgan Stanley Capital International market capitalization data;
- adjustments for stock market movements, exchange rate movements, and intercorporate cross-holdings.

The adjusted market values exclude intercorporate cross-holdings. This adjustment matters because some countries, especially Japan and Germany, had large corporate cross-shareholdings that would otherwise overstate freely held market capitalization.

**Interpretation.** The table is not just a count of market size. It attempts to measure the equity wealth actually available and relevant for portfolio allocation.

## 2.5 Domestic ownership shares

The paper reports that most corporate equity is held domestically. Approximate domestic ownership shares of the world's five largest stock markets are:

| Market                   | United States | Japan | United Kingdom | Germany | France |
|--------------------------|---------------|-------|----------------|---------|--------|
| Domestic ownership share | 92.2%         | 95.7% | 92.0%          | 79.0%   | 89.4%  |

**Interpretation.** The portfolio-weight evidence and the market-ownership evidence point in the same direction: most equity is owned by investors from the same country as the issuing firms.

## 2.6 Return data and covariance matrix

The paper studies real equity returns over 1975–1989. Returns are measured from the perspective of a U.S. investor, with exchange rate risk hedged by using three-month forward contracts to lock in exchange rates for the initial investment amount each quarter.

Let  $r_t$  be the vector of real returns across the six markets and let:

$$\Sigma = \text{Cov}(r_t). \quad (5)$$

The average pairwise correlation across the six markets is:

$$\bar{\rho} \approx 0.502. \quad (6)$$

The paper notes that the correlations are similar whether returns are measured in dollars, yen, or pounds, and whether exchange-rate risk is hedged or unhedged.

**Interpretation.** A correlation of 0.502 is high enough that national markets move together, but far below one. This creates potential risk reduction from holding foreign equities.

## 2.7 Value-weighted benchmark portfolio

The value-weighted benchmark is based on adjusted market values:

$$w_i^V = \frac{MV_i}{\sum_{k \in \mathcal{C}} MV_k}. \quad (7)$$

Using the adjusted market values in Table 1, the approximate benchmark weights are:

| Market             | U.S.  | Japan | U.K.  | France | Germany | Canada |
|--------------------|-------|-------|-------|--------|---------|--------|
| Value-weight share | 47.6% | 26.4% | 13.8% | 4.3%   | 3.8%    | 3.8%   |

**Interpretation.** A globally diversified investor would hold a large U.S. position because the U.S. market is large, but not a 94% U.S. position. The difference between actual weights and value weights is the home-bias gap.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Economic mechanism

The paper asks what beliefs make the observed portfolio weights optimal. The core mechanism is:

$$\text{high domestic portfolio weight} \implies \text{high domestic expected return or low perceived domestic risk.} \quad (8)$$

Because the covariance matrix is estimated from data and the risk-aversion coefficient is calibrated, the observed portfolio weights can be inverted to recover implied expected returns.

**Interpretation.** The model is a diagnostic device. It does not prove that investors consciously hold a particular expected return vector; it shows what beliefs would be necessary for their observed choices to be optimal in a standard portfolio model.

### 3.2 Investor utility

The representative investor in each country has exponential utility over wealth:

$$U(W) = -\exp\left(-A \frac{W}{W_0}\right), \quad (9)$$

where:

- $W$  is terminal wealth;
- $W_0$  is initial wealth;

- $A$  is the risk-aversion parameter;
- the paper uses  $A = 3$ .

Assuming returns are approximately normal, maximizing expected utility is equivalent to a mean–variance portfolio problem:

$$\max_w w' \mu - \frac{A}{2} w' \Sigma w, \quad (10)$$

where  $\mu$  is the vector of expected real returns and  $\Sigma$  is the return covariance matrix.

**Interpretation.** The investor likes expected returns but dislikes portfolio variance. The strength of the dislike is governed by  $A$ .

### 3.3 First-order condition

The optimal portfolio weights satisfy:

$$\mu = A \Sigma w^*. \quad (11)$$

Equivalently, for a given portfolio  $w^*$ , the expected return vector that rationalizes that portfolio is:

$$\mu(w^*, \Sigma, A) = A \Sigma w^*. \quad (12)$$

**Interpretation.** The paper reverses the usual portfolio problem. Instead of asking, “given expected returns, what portfolio should investors hold?”, it asks, “given the portfolio investors actually hold, what expected returns must they believe?”

### 3.4 Actual portfolio-implied expected returns

For investors in country  $j$ , the observed portfolio vector is  $w_j^A$ , where  $A$  here denotes actual rather than risk aversion. The implied expected returns are:

$$\mu_j^A = A \Sigma w_j^A. \quad (13)$$

For example, if U.S. investors place 93.8% of their equity wealth in U.S. stocks, then the model asks what expected return on U.S. stocks makes that concentration optimal, given the covariance matrix.

**Interpretation.** A highly concentrated domestic portfolio generally requires a high domestic expected return relative to foreign expected returns.

### 3.5 Value-weighted portfolio-implied expected returns

The authors also compute the expected return vector that would make the global value-weighted portfolio optimal:

$$\mu^V = A\Sigma w^V. \quad (14)$$

Then they compare actual-portfolio implied returns with value-weighted implied returns:

$$d_j = \mu_j^A - \mu^V. \quad (15)$$

**Interpretation.** The deviation  $d_j$  is the belief wedge required to justify observed portfolio weights relative to a globally diversified benchmark. Positive values on the domestic market are interpreted as domestic optimism.

### 3.6 Home-bias optimism wedge

For investor country  $j$ , define the home-market wedge:

$$\text{HomeBias}_j = d_{j,j} = \mu_{j,j}^A - \mu_j^V. \quad (16)$$

The paper's calculations imply approximately:

$$\text{HomeBias}_{\text{US}} \approx 0.9\%, \quad \text{HomeBias}_{\text{Japan}} \approx 2.5\%, \quad \text{HomeBias}_{\text{UK}} \approx 4.4\%. \quad (17)$$

In basis points:

$$0.9\% = 90\text{bp}, \quad 2.5\% = 250\text{bp}, \quad 4.4\% = 440\text{bp}. \quad (18)$$

**Interpretation.** The home-bias wedge is especially large for British and Japanese investors. In the model, they must either expect very high domestic returns or perceive foreign assets as much less attractive.

### 3.7 Risk and diversification cost

The cost of incomplete diversification comes from the fact that international returns are imperfectly correlated. Portfolio variance is:

$$\text{Var}(w'r) = w'\Sigma w. \quad (19)$$

If  $w$  is concentrated in the domestic market, the investor loses the variance reduction available from combining imperfectly correlated equity markets.

**Interpretation.** The issue is not that domestic equity is necessarily bad. The issue is that holding almost only domestic equity leaves diversification gains on the table unless domestic assets offer compensating returns or lower perceived risk.



### 3.8 What the model does not identify

The portfolio inversion cannot distinguish among three mechanisms:

$$\begin{aligned} \text{observed home bias} \quad \Longleftrightarrow \quad & \text{high expected domestic returns} \\ & \text{low perceived domestic risk} \\ & \text{institutional or psychological cost of foreign investing.} \end{aligned} \tag{20}$$

**Interpretation.** The model quantifies the size of the wedge. It does not by itself prove whether the wedge comes from rational information, irrational optimism, ambiguity, taxes, regulation, or familiarity.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 Nature of identification

This is not a causal identification paper. There is no treatment, no instrumental variable, and no regression coefficient whose causal interpretation depends on a quasi-experiment.

Instead, the paper uses a **revealed-preference calibration**:

$$\text{observed portfolios} + \text{estimated covariance matrix} + \text{risk aversion} \Rightarrow \text{implied expected returns.} \tag{21}$$

**Interpretation.** Credibility comes from the transparency of the calculation and the starkness of the resulting implied beliefs.

### 4.2 Main identifying assumptions

The revealed-preference calculation relies on four central assumptions:

1. **Mean–variance/exponential-utility behavior.** Investors behave as if they maximize expected utility with a tractable variance penalty.
2. **Common covariance matrix.** The return covariance matrix estimated from 1975–1989 returns is a reasonable approximation to the risk environment investors face.
3. **Risk aversion calibration.** A risk-aversion parameter of  $A = 3$  is a reasonable benchmark.
4. **Six-market universe.** Investors allocate equity wealth across the six major national equity markets considered in the paper.

**Interpretation.** If any of these assumptions changes, the exact basis-point numbers change. But the qualitative puzzle remains: observed portfolios are much more domestic than a standard diversification model predicts.

### 4.3 Why institutional constraints are unlikely to fully explain the evidence

The paper considers several institutional explanations:

- **Capital controls.** These mattered historically, especially before the liberalization of U.K. capital controls under Margaret Thatcher, but they are unlikely to explain the late-1980s portfolio patterns in full.
- **Taxes.** Dividend withholding taxes on foreign shareholders may reduce foreign returns, but for many taxable investors these payments can be credited against home-country taxes.
- **Transaction costs.** Trading costs may differ across markets, but transaction costs should push all investors toward low-cost markets rather than toward each investor's own domestic market.
- **Explicit limits.** Some restrictions existed, such as limits on foreign ownership or insurance-company foreign assets, but observed foreign holdings often appear below these limits.

**Interpretation.** Institutional barriers may contribute to home bias, but the paper argues that they are too small or too nonbinding to rationalize the observed degree of domestic concentration.

### 4.4 Evidence against binding constraints

The authors emphasize that cross-border equity holdings sometimes fall rather than rise toward regulatory ceilings. Examples include:

- foreign investors were substantial net sellers of Japanese shares in the mid-1980s;
- foreigners were net sellers of U.S. equities in 1988;
- gross cross-border equity flows were large relative to net flows.

**Interpretation.** If regulations were the primary binding force preventing diversification, one would expect investors to keep pushing up against those constraints. The observed net selling episodes suggest that investor choices, not just constraints, drive the low foreign allocation.

### 4.5 Behavioral and informational explanations

The paper highlights two behavioral/informational channels.

First, return expectations may differ across investor groups. For example, survey evidence from Shiller, Kon-ya, and Tsutsui shows that U.S. and Japanese portfolio managers had very different views about U.S. and Japanese stock returns in early 1990.

Second, investors may perceive familiar domestic assets as less risky than unfamiliar foreign assets:

$$\text{foreign asset unfamiliarity} \Rightarrow \text{higher perceived risk} \Rightarrow \text{lower foreign portfolio weight.} \quad (22)$$

**Interpretation.** Home bias may reflect a combination of optimism about the home market and ambiguity or competence effects regarding foreign markets.

## 4.6 Why mistaken beliefs can persist

The paper notes that expected returns are statistically difficult to estimate. Even using 60 years of U.S. stock market data, the standard error of the estimated mean annual return is about 200 basis points. A 95% confidence interval therefore spans about 800 basis points.

**Interpretation.** Because long-run expected equity returns are hard to learn precisely, investors can maintain heterogeneous beliefs about domestic versus foreign returns for a long time without being clearly contradicted by the data.

## 4.7 Relation to broader diversification puzzles

The paper connects international equity home bias to other underdiversification facts:

- individual investors often hold only a small number of stocks;
- households concentrate wealth in local residential real estate;
- human capital and housing exposure are often geographically correlated.

**Interpretation.** The international home-bias puzzle is one example of a broader tendency for investors to hold familiar, local, or domestic assets even when diversification theory suggests otherwise.

# 5 Tables and figures

## 5.1 Table 1: Equity portfolio weights

TABLE 1—EQUITY PORTFOLIO WEIGHTS:  
BRITISH, JAPANESE, U.S. INVESTORS

|         | Portfolio Weight |       |      | Adj.<br>Market<br>Value |
|---------|------------------|-------|------|-------------------------|
|         | U.S.             | Japan | U.K. |                         |
| U.S.    | .938             | .0131 | .059 | \$2941.3                |
| Japan   | .031             | .9811 | .048 | 1632.9                  |
| U.K.    | .011             | .0019 | .820 | 849.8                   |
| France  | .005             | .0013 | .032 | 265.4                   |
| Germany | .005             | .0013 | .035 | 235.8                   |
| Canada  | .010             | .0012 | .006 | 233.5                   |

*Note:* Estimates correspond to portfolio holdings in December, 1989. They are based on the authors' tabulations using data from the *U.S. Treasury Bulletin* and Michael Howell and Angela Cozzini (1990). Adjusted market values exclude intercorporate cross-holdings from total market value, and correspond to June 1990 values.

**Figure 1:** Equity portfolio weights for U.S., Japanese, and British investors.

Read.

- U.S. investors hold 93.8% of their six-market equity portfolio in U.S. stocks.
- Japanese investors hold 98.11% of their portfolio in Japanese stocks.
- British investors hold 82.0% of their portfolio in U.K. stocks.
- The United States has the largest adjusted market value, followed by Japan and the United Kingdom.
- Even though the U.S. market is large, its value-weighted share of the six-market universe is far below the 93.8% U.S. portfolio weight held by U.S. investors.

### Economic message.

Table 1 is the central descriptive fact. Investors do not hold the world portfolio; they hold overwhelmingly domestic equity portfolios. The table turns “home bias” from a vague idea into a portfolio-weight fact.

## 5.2 Table 2: Expected real returns implied by actual portfolio holdings

| TABLE 2—EXPECTED REAL RETURNS<br>IMPLIED BY ACTUAL PORTFOLIO HOLDINGS                |      |       |      |
|--------------------------------------------------------------------------------------|------|-------|------|
|                                                                                      | U.S. | Japan | U.K. |
| <b>A. Expected Returns Needed to Justify Observed Portfolio Weights</b>              |      |       |      |
| U.S.                                                                                 | 5.5  | 3.1   | 4.4  |
| Japan                                                                                | 3.2  | 6.6   | 3.8  |
| U.K.                                                                                 | 4.5  | 3.8   | 9.6  |
| France                                                                               | 4.3  | 3.4   | 5.3  |
| Germany                                                                              | 3.6  | 3.0   | 4.8  |
| Canada                                                                               | 4.7  | 3.0   | 4.0  |
| <b>B. Deviation Between Implied Returns for Actual and Value-Weighted Portfolios</b> |      |       |      |
| U.S.                                                                                 | 0.9  | −1.5  | −0.2 |
| Japan                                                                                | −1.1 | 2.5   | −0.3 |
| U.K.                                                                                 | −0.7 | −1.4  | 4.4  |
| France                                                                               | −0.3 | −1.2  | 0.7  |
| Germany                                                                              | −0.2 | −0.8  | 1.0  |
| Canada                                                                               | 0.5  | −1.2  | −0.2 |

*Note:* See text for further description of calculations.

**Figure 2:** Expected real returns implied by observed and value-weighted portfolios.

Read.

- Panel A reports the expected real returns required to make observed portfolio weights optimal.
- U.S. investors’ observed portfolio implies a 5.5% expected return on U.S. equities, compared with 3.2% on Japanese equities and 4.5% on U.K. equities.

- Japanese investors' observed portfolio implies a 6.6% expected return on Japanese equities, compared with 3.1% on U.S. equities.
- British investors' observed portfolio implies a 9.6% expected return on U.K. equities, compared with 4.4% on U.S. equities and 3.8% on Japanese equities.
- Panel B subtracts the implied returns that would support the global value-weighted portfolio. The domestic wedges are about +0.9% for U.S. investors, +2.5% for Japanese investors, and +4.4% for British investors.

#### **Economic message.**

Table 2 is the main quantitative contribution. The observed home-biased portfolios require large domestic expected-return advantages. These wedges are too large to dismiss as a rounding error in portfolio construction.

### **5.3 Summary exhibit: what must rationalize home bias?**

| Candidate explanation             | How it rationalizes domestic concentration                                | Paper's assessment                                                                   |
|-----------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Higher domestic expected returns  | Investors believe home equities will outperform foreign equities          | Quantitatively consistent with observed portfolios but implies large optimism wedges |
| Lower perceived domestic risk     | Investors view domestic markets as safer or more familiar                 | Plausible behavioral channel; difficult to measure directly                          |
| Taxes                             | Foreign dividends face withholding taxes or different after-tax treatment | Too small for many investors to explain the full puzzle                              |
| Transaction costs                 | Foreign equity trading is more expensive                                  | Cannot explain why every country tilts toward its own market                         |
| Capital controls or formal limits | Investors are prevented from holding enough foreign equity                | Historically relevant, but late-1980s holdings appear below many constraints         |
| Information and familiarity       | Investors know domestic firms, institutions, and markets better           | Plausible explanation for persistent home bias                                       |

**Economic message.** The paper's preferred interpretation is not that institutions are irrelevant. Rather, the magnitude and direction of portfolio tilts suggest that investor beliefs, information, and familiarity are central.

## 5.4 Summary exhibit: model logic

| Step                            | Logic                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------|
| 1. Observe portfolio weights    | Investors in each country allocate most equity wealth to domestic stocks.                        |
| 2. Estimate covariance matrix   | International equity returns are imperfectly correlated, so diversification is valuable.         |
| 3. Calibrate risk aversion      | Use $A = 3$ as a benchmark risk-aversion parameter.                                              |
| 4. Invert first-order condition | Compute $\mu = A\Sigma w$ for each observed portfolio.                                           |
| 5. Compare with value weights   | Ask how much domestic optimism is needed relative to a globally diversified benchmark.           |
| 6. Interpret wedge              | Large domestic wedges imply home bias is hard to reconcile with standard diversification theory. |

**Economic message.** The paper's strength is its simplicity. A short inversion of portfolio first-order conditions reveals that home bias carries strong implications for expected returns or perceived risks.

## 6 Short summary

- **Question.** Why do investors hold so much domestic equity despite the risk-reduction benefits of international diversification?
- **Data.** The paper uses late-1980s estimates of U.S., Japanese, and British portfolio weights across six major equity markets, together with 1975–1989 international equity return covariances.
- **Core fact.** U.S., Japanese, and British investors hold roughly 94%, 98%, and 82% of their equity portfolios in domestic stocks, respectively.
- **Model.** A representative investor with exponential utility and risk aversion  $A = 3$  chooses portfolio weights according to  $\mu = A\Sigma w$ .
- **Main result.** The observed domestic concentration implies large domestic expected-return advantages, especially for Japanese and British investors.
- **Benchmark result.** Relative to a global value-weighted portfolio, the required domestic optimism is about 90 basis points for U.S. investors, 250 basis points for Japanese investors, and 440 basis points for British investors.
- **Institutional explanations.** Taxes, transaction costs, capital controls, and explicit limits may matter, but they are unlikely to fully explain the magnitude and country-specific direction of home bias.
- **Behavioral explanation.** Domestic optimism, familiarity, and perceived risk of foreign assets are plausible explanations.
- **Broader message.** The international home-bias puzzle is part of a wider pattern of underdiversification in household and institutional portfolios.

## 7 Conclusion

### 7.1 Core conclusion

French and Poterba show that international equity portfolios in the late 1980s are far more domestic than standard diversification logic predicts. Using a simple portfolio model, they translate observed domestic concentration into the expected return beliefs that would be needed to rationalize it. The required belief wedges are large: investors must view their own domestic markets as substantially more attractive than foreign markets.

### 7.2 Economic intuition

If national stock markets are imperfectly correlated, a diversified international portfolio should reduce risk. Therefore, an investor who holds almost entirely domestic stocks must have a reason. In the model, that reason must appear as high expected domestic returns, low perceived domestic risk, or a cost/constraint associated with foreign investing. Because the institutional explanations seem too weak to fully account for the evidence, beliefs and familiarity become natural explanations.

### 7.3 Takeaway

The paper is an early and influential statement of the equity home-bias puzzle. Its main lesson is that portfolio weights are informative: when investors overwhelmingly hold domestic assets, their choices imply strong beliefs or frictions. International diversification is financially valuable, but investors appear reluctant to exploit it fully.